

- 6 Insects and other invertebrates are a good source of protein, unsaturated fatty acids, iron and other nutrients.

Recent research has shown that some insects and other invertebrates have a high antioxidant content.

(a) How many of the following statements are true for proteins?

- Proteins contain only the elements carbon, hydrogen and nitrogen.
- Proteins can be made of more than one polypeptide chain.
- Proteins contain hydrogen bonds.

(1)

- ☐ **A** 0
- ☐ **B** 1
- ☐ **C** 2
- ☐ **D** 3

(b) Which row of the table is correct for an unsaturated fatty acid?

(1)

	Contains carbon–carbon double bonds	Carbon : hydrogen ratio compared with a saturated fatty acid is
☐ A	no	lower
☐ B	no	higher
☐ C	yes	lower
☐ D	yes	higher

(c) Name **one** organic molecule that contains iron.

(1)

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(d) Explain why dietary antioxidants are important.

(4)

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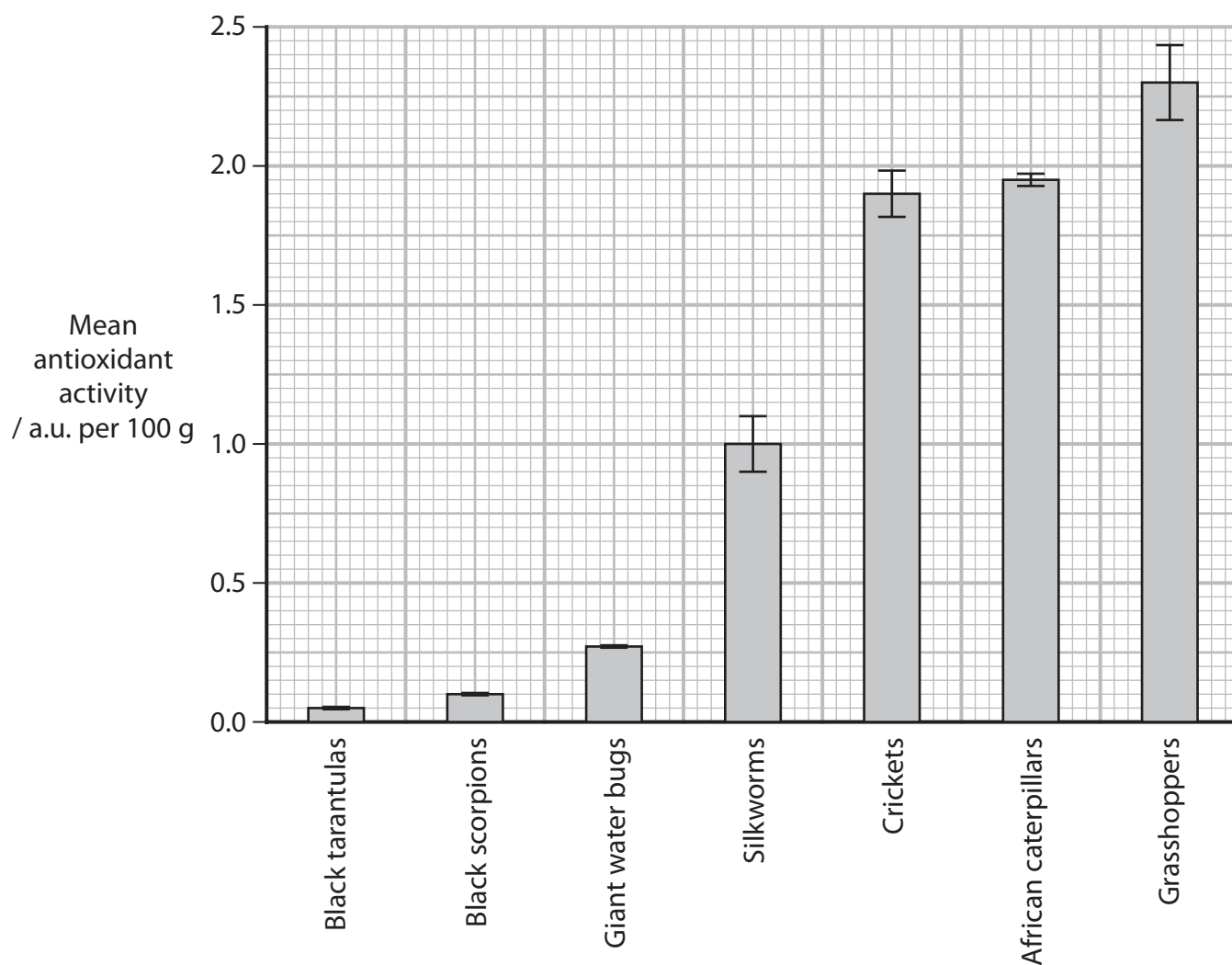


(e) The antioxidant activity of some insects and arachnids was investigated.

The table shows the dietary habits of the organisms used in this investigation.

Name of organism	Type of organism	Dietary habits
Black tarantulas	arachnid	eat animals
Black scorpions	arachnid	eat animals
Giant water bugs	insect	eat animals
Silkworms	insect	eat plants
Crickets	insect	eat plants
African caterpillars	insect	eat plants
Grasshoppers	insect	eat plants

The graph shows the mean antioxidant activity of extracts made from these organisms.



- (i) The antioxidant activity of orange juice is 0.80 a.u. per 100 g.

Calculate how many times more antioxidant activity there is in 100 g of grasshopper extract.

Give your answer to two significant figures.

(1)

Answer

- (ii) Calculate the percentage difference between the antioxidant activity in black scorpions and crickets.

(1)

Answer%

- (iii) Comment on the results of this investigation.

Use the information in the table and the graph to support your answer.

(3)

(Total for Question 6 = 12 marks)

